# L5-6: Quantum Computing: Foundation, Applications

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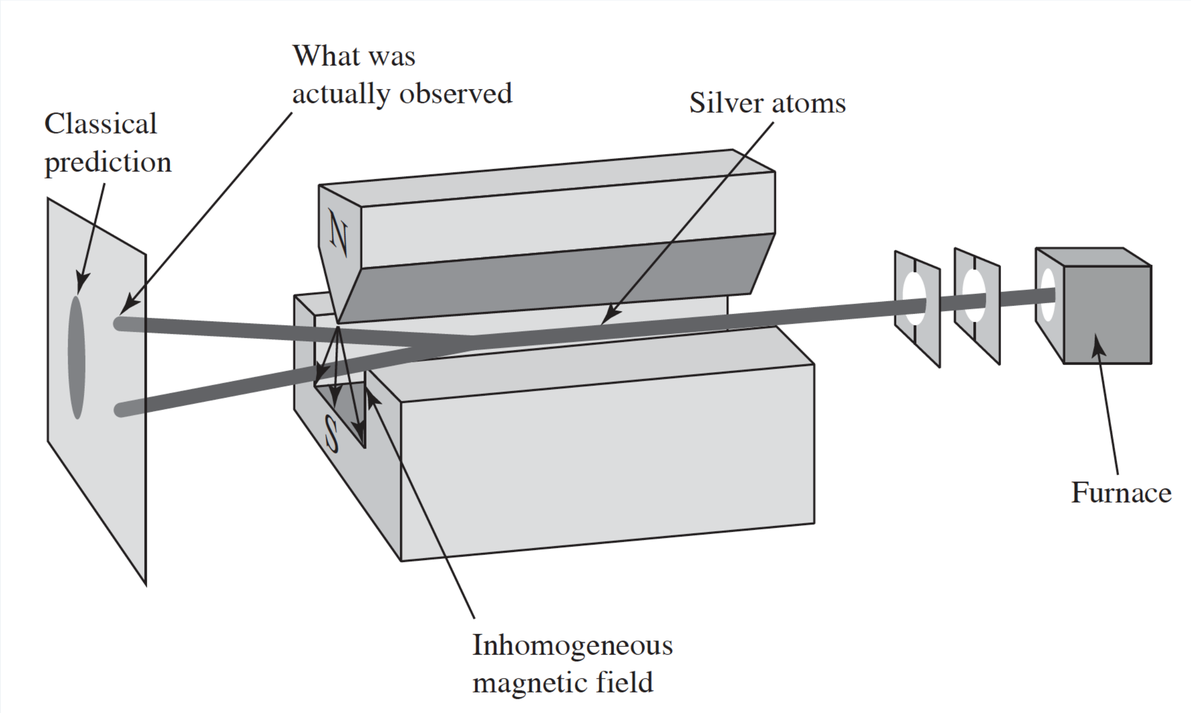
# This session

**Foundations of the “Quantum” Behavior**

**Applications of Quantum Computing**

**Postulates of Quantum Mechanics**

# QM Foundation: The Stern-Gerlach Experiment takeaway




**Measuring along a new axis doesn't just reveal new information — it rewrites the system's state and erases incompatible previous information.**

J. J. Sakurai, Modern Quantum Mechanics

## The Stern–Gerlach experiment shows:

- **Angular momentum is quantized.**
- **Measurement outcomes are discrete.**
- **Spin-1/2 systems are 2D complex vector spaces (?)**
- **Changing measurement axis reveals superposition.**
- **This is the physical origin of the qubit!**

## From this experiment, we get:

- **The idea of Superposition**
- **Basis change, Unitary rotations, State Collapse**
- **Measurement is not passive observation ⟹It actively prepares the system in a new state.**
- **Measurement of conjugate variables destroy the information!**

# Quantum Computing

**Harnesses the quantum behavior for computing!**

**Fundamentally different from classical computing paradigm**

Hardware, Algorithms, Architecture

# Applications of Quantum Computing

## Pharmaceuticals & Drug Discovery

- Quantum computers can simulate the behavior of matter at the molecular level.
- It is impossible for classical computers.
- **Protein Folding:** Accurately predicting how proteins fold to identify the root causes of various diseases
- **Molecular Docking:** Simulating how new drug compounds interact with human cells to reduce the long (sometimes 10+ years) drug-development cycle.

## Cybersecurity & the “Q-Day” Readiness

- Quantum computing poses a dual threat and opportunity for global security.
- **Breaking Encryption:** Shor’s Algorithm could potentially crack current RSA and ECC encryption.
- **Post-Quantum Cryptography (PQC):** Developing new “quantum-safe” mathematical frameworks to protect data.
- **Quantum Key Distribution (QKD):** Using the principle of entanglement to create unhackable communication channels.

# Applications of Quantum Computing

## Material Science & Energy

- By modeling chemical reactions at a quantum level, we can design materials with specific properties.
- **Next-Gen Batteries:** Developing high-capacity, faster-charging solid-state batteries for EVs (?)
- **Nitrogen Fixation:** Improving the Haber-Bosch process to create more efficient fertilizers, potentially reducing global energy consumption by 1-2%.
- **Carbon Capture:** Designing new catalysts that can efficiently "scrub" $CO_2$ from the atmosphere.

## Finance & Risk Management

- **Many financial problems reduce to high-dimensional sampling and optimization tasks, where certain quantum algorithms provide provable speedups over classical ones.**
- **Portfolio Optimization:** Finding the "efficient frontier" to maximize returns while minimizing risk across thousands of assets.
- **Fraud Detection:** Using Quantum Machine Learning (QML) to identify subtle patterns in transaction data that indicate money laundering or credit card fraud.

# Applications of Quantum Computing

## Logistics & Supply Chain

- Solving “combinatorial” problems where the number of possible solutions exceeds the number of atoms in the universe.
- Route Optimization: Solving the “Traveling Salesperson Problem” for global shipping fleets to minimize fuel and time.
- Traffic Management: Optimizing urban traffic flow in real-time to eliminate congestion in “Smart Cities.”

## Quantum-Enhanced AI

- Quantum computers can potentially act as “accelerators” for the most demanding parts of the AI pipeline.
- Faster Training: Reducing the time to train Large Language Models (LLMs) from months to hours.
- Complex Pattern Recognition: Enabling AI to “see” correlations in high-dimensional data that classical hardware misses.

**Key Takeaway:**

- **We are currently in the NISQ Era (Noisy Intermediate-Scale Quantum). We aren't at “Fault Tolerant Quantum Computing”.**
- **Yet, hybrid workflows where classical and quantum computers work together, are already looking very promising in 2026!**

# Quantum Computing: Critical Challenges

**Building a practical quantum computer is a transition from theoretical physics to complex systems engineering.**

## The “Noise” Problem (Decoherence)

- **Quantum states are extremely fragile:** Even tiny environmental interactions (heat, EM noise, cosmic rays) destroy superposition in microseconds.
- **Effect:** Decoherence causes computations to collapse into random classical noise before completion.
- **Current solution:** Operate at ultra-low temperatures (~0.015 K) with heavy vacuum and electromagnetic shielding.

## The Scalability Paradox (QEC)

- **We are in the NISQ era:** Current quantum computers are noisy; progress requires higher-quality qubits, not just more qubits.
- **Logical vs Physical qubits:** Due to high error rates, many physical qubits are combined using Quantum Error Correction (QEC) to form one stable logical qubit.
- **Scaling challenge:** 100 to 1,000 physical qubits are needed for 1 logical qubit. Practical large-scale algorithms would require millions of physical qubits.

## The “Wiring” & Control Bottleneck

- **Scaling hardware is an engineering nightmare:** Each qubit needs its own control wiring (“forest of wires”); scaling to millions would require impractically massive cryogenic infrastructure.
- **Cryogenic control challenge:** Classical electronics must operate at ultra-low temperatures to reduce latency and heat, requiring new cryo-compatible control systems.

# Foundation: Postulates of Quantum Mechanics

- A system is represented by a wavefunction
  - Hilbert space
- Evolution is governed by Schrödinger equation
  - Hamiltonian
- Measurement
  - Hermitian operators
- **Superposition**
- **Entanglement**

**Interference**

**For a spin-½ qubit** $\mathcal{H} = \mathbb{C}^2$

**Basis states:** $\{|\uparrow\rangle, |\downarrow\rangle\}$

$$|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

**Hermitian Operators (Observables)** $\hat{S}_i = \frac{\hbar}{2}\sigma_i$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

**General Hermitian Operator:** $\hat{O} = a_0\mathbb{I} + \vec{a} \cdot \vec{\sigma}$

**Hamiltonian of a Two-Level Spin Qubit** $\hat{H} = \frac{1}{2}\hbar\vec{\Omega} \cdot \vec{\sigma}$

Thank you